COASTAL ENGINEERING – CIVE70004

COURSEWORK

AIMS AND LEARNING OUTCOMES

The aim of this coursework exercise is to practise key calculations required in Coastal Engineering. These involve the selection of an appropriate method to calculate wave kinematics. In practice, this type of analysis is commonly carried out in coastal locations using data from wave buoys and is used to estimate the design conditions for coastal structures.

TASKS

A wave buoy has been deployed at a water depth of d=30 m off the east coast of the UK. The wave buoy is used to assess the wave climate in the area and establish the design criteria for an offshore wind farm and coastal protection structures closer to the shore.

During a large winter storm, the largest wave recorded by this buoy is shown in Figure 1. Using this wave record, calculate the depth variation in the horizontal wave-induced velocity, $u(z)$, arising beneath the largest wave crest. Determine the maximum velocity arising at any elevation. Comment on the appropriateness of the various kinematic models applied and make suggestions as to which model provides the best design solution. **Note:** The amplitude and phase spectrum relating to this wave are provided on Figure 2. Separate Matlab matfiles containing the data given in Figures 1 and 2 has been uploaded to Blackboard.

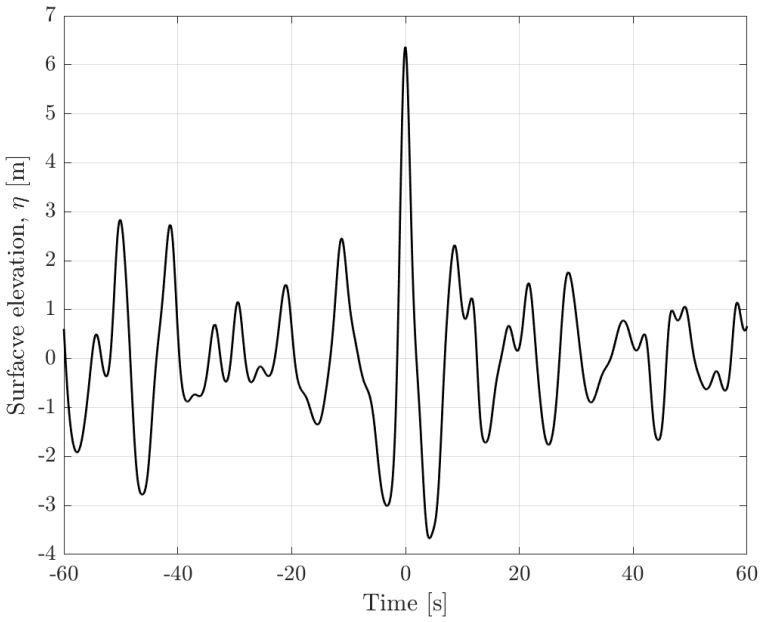


Figure 1: Time-history of the largest wave at d=30 m.

ASSESSMENT

The final report should indicate clearly and concisely the methods of calculation. Your final comments should be limited to 3 sides of A4 paper.

Due date: Monday 23rd February at 9 am

ADDITIONAL MATERIAL

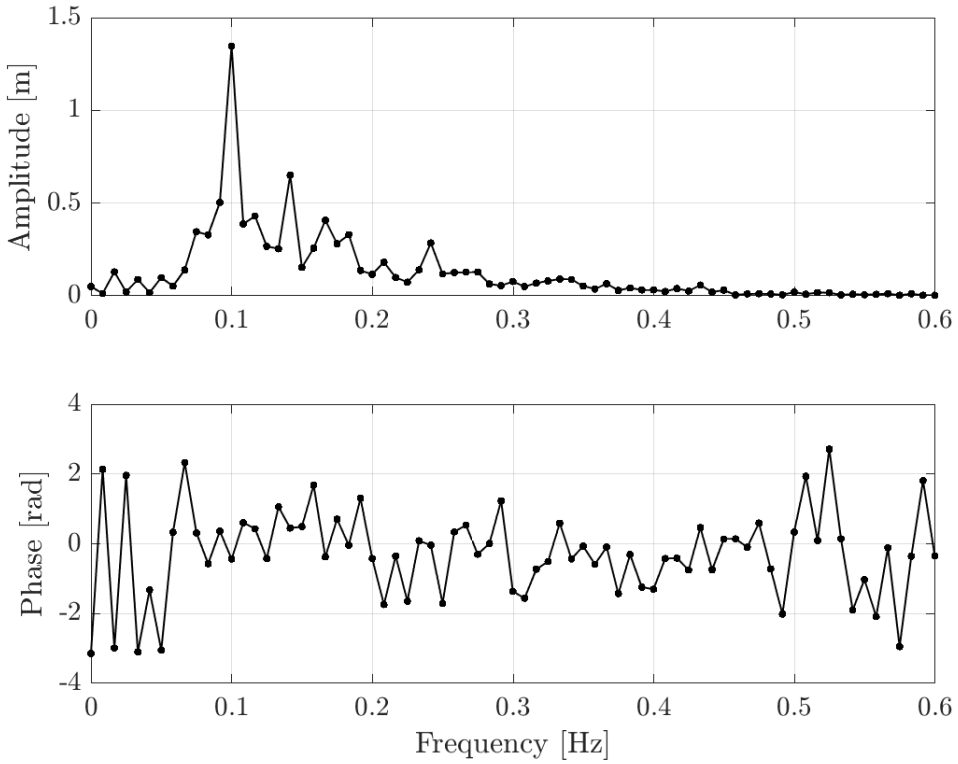


Figure 2: Amplitude and Phase spectra corresponding to the largest wave shown of Figure 1.